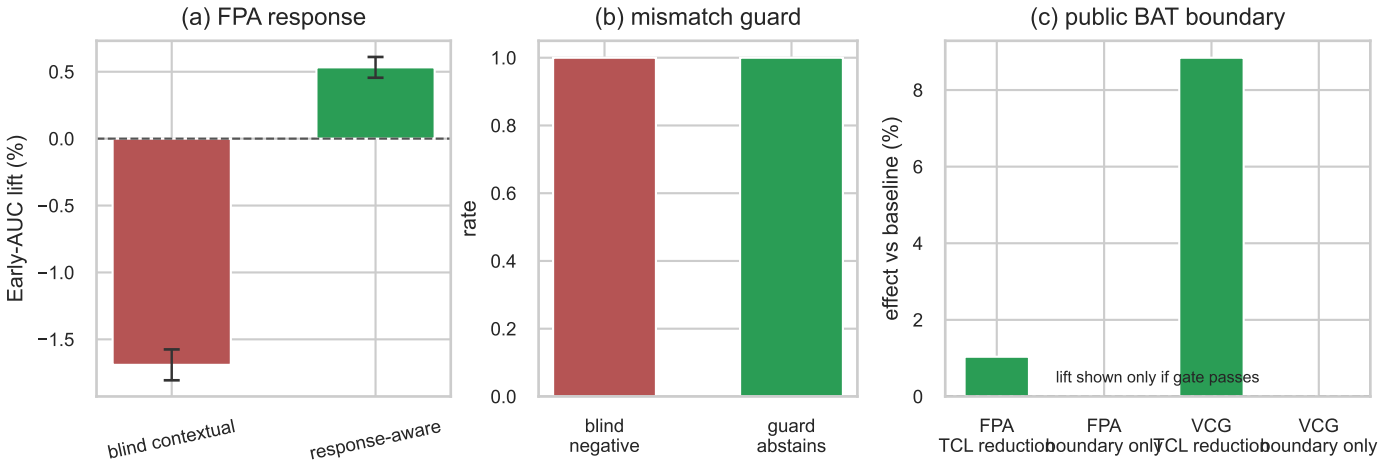
(a) FPA response
Early-AUC lift (%)
0.5
0.0
−0.5
−1.0
−1.5
blind contextual
response-aware
(b) mismatch guard
rate
1.0
0.8
0.6
0.4
0.2
0.0
blind
negative
guard
abstains
(c) public BAT boundary
effect vs baseline (%)
8
6
4
2
0
lift shown only if gate passes
FPA
TCL reduction
FPA
boundary only
VCG
TCL reduction
VCG
boundary only